DATA ANALYTICS LEVEL III

Day 7: SQL Filtering & Sorting Practice — Hospital Patients Dataset

Student Activities

Day 7 — Student Activities Overview

Complete the following activities during your asynchronous session (3.5 hours total).

Unit of Competency: Prepare Data Sets (Core 2)

Learning Outcomes Assessed: LO3 — Filter data; LO4 — Sort and arrange data

Activity	Type	Duration	Weight
SQL Filter & Sort Exercises	Formative	1.5 hours	Checked for completion
Real-World Hospital Query Scenarios	Formative	0.75 hours	Graded (40 pts)
Reading: SQL Filtering Deep Dive	Self-study	0.75 hours	N/A
Learning Journal Entry #7	Reflective	0.5 hours	Part of participation grade

Dataset Reference

File Needed: Day07_practice.db (download from LMS)

All activities below use a single table called patients (50 records) inside Day07_practice.db. The table represents hospital admissions for the year 2025.

Table: patients (50 rows)

Column	Type	Description
`patient_id`	INTEGER	Unique patient ID (1–50)
`patient_name`	TEXT	Full name of the patient
`age`	INTEGER	Age in years (5–72)

gender	TEXT	'Male' or 'Female'
blood_type	TEXT	Blood type (e.g., 'O+', 'A-', 'AB+')
city	TEXT	City of residence
department	TEXT	Hospital department (Cardiology, Pediatrics, Oncology, etc.)
diagnosis	TEXT	Primary diagnosis
doctor_name	TEXT	Attending physician
admission_date	TEXT	Admission date (YYYY-MM-DD format)
room_type	TEXT	'Private', 'Semi-Private', 'Ward', or 'ICU'
billing_amount	REAL	Total bill in Philippine pesos (₱)
status	TEXT	'Admitted', 'Discharged', 'Critical', or 'Stable'

Activity 1: SQL Filter & Sort Exercises

Type: Formative (Practice)

Time: 1.5 hours

File Needed: Day07_practice.db

Using VS Code + SQLTools connected to Day07_practice.db, write SQL queries to answer each question. Record your queries and results.

Part A: AND, OR, NOT (5 queries)

1. Show all patients from the Cardiology department where age is greater than 50.
2. Show all patients who live in Cebu City OR Davao City.
3. Show all patients who are NOT in the Pediatrics department.
4. Show all Female patients whose status is 'Critical'.
5. Show all patients where the room_type is 'ICU' AND billing_amount is greater than ₱150,000.

Part B: BETWEEN, IN, LIKE (5 queries)

6. Show all patients whose age is BETWEEN 18 AND 35.
7. Show all patients admitted in Q1 2025 (admission_date BETWEEN '2025-01-01' AND '2025-03-31').
8. Show all patients where the department is IN ('Cardiology', 'Oncology', 'Neurology').
9. Show all patients whose diagnosis starts with the word 'Heart' (use LIKE).
10. Show all patients whose doctor_name contains 'Tan' anywhere in the name.

Part C: ORDER BY and LIMIT (5 queries)

11. Show all patients sorted by billing_amount from highest to lowest (DESC).
12. Show the TOP 5 most expensive admissions (highest billing_amount). Use LIMIT.
13. Show all Pediatrics patients sorted by admission_date oldest first (ASC).
14. Show the 10 most recent admissions (sort by admission_date newest first, then LIMIT 10).
15. Show all patients from Metro Manila cities (Manila, Makati, Quezon City, Pasig), sorted by department A–Z, then by billing_amount highest first.

Submission Format

For each query, write the question number, your SQL, and how many rows returned. Example:

```
-- Part A, Question 1: Cardiology patients older than 50
SELECT * FROM patients
WHERE department = 'Cardiology' AND age > 50;
-- Result: X rows returned
```

Save as: Day07_SQL_Exercises_[YourName].sql or .txt

Activity 2: Real-World Hospital Query Scenarios

Type: Graded (40 points)

Time: 45 minutes

File Needed: Day07_practice.db

You are a junior data analyst working for the hospital administration office. The medical director and department heads send you 5 questions. Write the SQL query that answers each one, and add a 1–2 sentence interpretation of the results.

Scenario 1: ICU Critical Care Report (8 points)

The medical director needs an urgent report of all patients currently in the ICU whose status is 'Critical'. Show their patient_name, age, diagnosis, doctor_name, and billing_amount. Sort by billing_amount from highest to lowest.

Write your query and interpretation.

Scenario 2: Senior Citizen Care Analysis (8 points)

The Geriatrics committee is studying admissions of senior patients. Show all patients whose age is BETWEEN 60 AND 75 AND whose department is in ('Cardiology', 'Oncology', 'Neurology'). Show patient_name, age, department, diagnosis, and status. Sort by age (oldest first).

Write your query and interpretation.

Scenario 3: High-Value Billing Review (8 points)

The finance department needs to flag high-value admissions for insurance audit. Show all patients whose billing_amount is BETWEEN ₱100,000 AND ₱250,000 AND whose room_type is 'Private' OR 'ICU'. Show patient_name, room_type, billing_amount, and department. Sort by billing_amount (highest first), then LIMIT the results to the TOP 10.

Write your query and interpretation.

Scenario 4: Diagnosis Pattern Search (8 points)

A research team is studying cancer cases across the hospital. Search the patients table for all patients whose diagnosis contains the word 'Cancer' (use LIKE). Show patient_name, age, gender, diagnosis, and billing_amount. Sort the results by age (youngest first).

Write your query and interpretation.

Scenario 5: Recent Admissions Dashboard (8 points)

The front-desk supervisor wants a quick dashboard of the most recent admissions from Q3 2025 (July 1 – September 30). Show all patients admitted in this period whose status is 'Discharged' OR 'Stable'. Display patient_name, admission_date, department, room_type, and status. Sort by admission_date (newest first), then LIMIT to 15 records.

Write your query and interpretation.

Grading Rubric

Component	Points
Correct SQL query (syntax, logic, correct table/columns)	5 pts per scenario
Meaningful interpretation (1–2 sentences explaining the results)	3 pts per scenario
Total	**40 points**

Save as: Day07_Hospital_Queries_[YourName].sql or .txt

Learning Journal Entry #7

Time: 30 minutes

Reflect on today's learning experience. Write at least 5 sentences addressing these prompts:

16. How comfortable are you now with combining AND, OR, and NOT in a single WHERE clause? Which combination was the trickiest?
17. Compare BETWEEN, IN, and LIKE — which one feels most useful for hospital data, and why?

18. Why does ORDER BY almost always come BEFORE LIMIT in a query? What happens if the order is wrong?
19. If a doctor asked you for the "5 most expensive cases this month", exactly which clauses would you use? Walk through your thinking.
20. What is one real-world hospital question you would want to answer with SQL filtering/sorting in a future job?

Save as: Day07_Journal_[YourName].docx

Upload all submissions to LMS before the next session.

Summary of Submissions

Submission	Filename	Due
SQL Filter & Sort Exercises	Day07_SQL_Exercises_[YourName].sql	Before next session
Hospital Query Scenarios	Day07_Hospital_Queries_[YourName].sql	Before next session
Learning Journal #7	Day07_Journal_[YourName].docx	Before next session